KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRICAL ENGINEERING THEORY

DATE: 25th February 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

ELECTRICAL ENGINEERING TECHNOLOGY

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Which of the following statements is false?
   a) A DC motor converts electrical energy to mechanical energy
   b) The efficiency of a DC motor is the ratio output power to input power
   c) A DC generator converts mechanical power to electrical power
   d) The efficiency of a DC generator is the ratio of input power to output power
2. In a DC generator, iron losses are made up of
   a) Hysteresis and frictional loses
   b) Hysteresis, eddy currents, brush and contact losses
   c) Hysteresis and eddy current losses
   d) Hysteresis, eddy current losses and copper losses
3. In a DC motor, the stator resistor
   a) Limits armature current to a safe starting value
   b) Controls the speed of the machine
   c) Prevents field current from flowing through the machine
   d) Limits other field current to a scale starting value
4. Laminations of a core are generally made of
   a) Cast iron
   b) Carbon
   c) Silicon steel
   d) Stainless steel
5. The commutator segments are connected to the armature conductors by means of
   a) Copper lugs
   b) Resistance
   c) Insulation pads
   d) Brazing
6. A welding generator will have
   a) Lap winding
   b) Wave winding
   c) Delta winding
   d) Duplex wave winding
7. In a DC machine armature reaction is produced by
   a) Field current
   b) Armature conductors
   c) Field pole winding
   d) Armature current
8. Which one of the following is the rotating part of an electrical machine?
   a) Yoke
   b) Brushes
   c) Commutator
   d) Field poles

9. The speed of DC motor may be increases by
   a) Increasing the armature current
   b) Decreasing the field current
   c) Decreasing the applied voltage
   d) Increasing the filed current
10. The armature torque of a DC motor is a function of its
   a) Field for alone
   b) Armature current alone
   c) Speed alone
   d) Both field flux and armature current
11. DC motors are considered suitable for their application in
   a) Fans
   b) Water pumps
   c) Traction
   d) Floor mills
12. Which one of the following is the primary reason of placing field magnets on the rotor of an alternator?
   a) Small power in the stator
   b) Insulation of high-voltage is made easy on the stator than rotor
   c) Large power on the stator
   d) Large current on the stator
13. If the diameter and length of the stator of a high speed turbo alternator and low speed hydro generator of identical ratings are compared
   a) Turbo-generator has a larger diameter and larger length
   b) Turbo-generator has a smaller diameter and larger length
   c) The turbo generator has a larger diameter and small length
   d) The turbo generator has a smaller diameter and larger lengths
14. Synchronous reactance of an alternator represents
   a) Armature reactance and leakage reactance
   b) A reactance operating at synchronous speed
   c) Field winding reactance
   d) A reactance connected in series with capacitance of the generator
15. An electric motor voltage in which both the rotor and station field rotates with the same speed is called_____
   a) Pc
   b) Induction
   c) Synchronous
   d) Universal
16. The direction of rotation of a synchronous motor can be reversed by?
   a) Reversing current to the field winding
   b) Reversing supply phase sequence
   c) Reversing polarity of the rotor poles
   d) None of the above

17. An induction motor works with
    a) DC only
    b) AC only
    c) Both AC and DC
    d) None of the above
18. The rotor of a three phase wound induction motor is provided with
    a) Single phase winding
    b) Three phase winding
    c) Heavy copper and copper aluminum bars placed in rotor slots
    d) Heavy short circuited end rings
19. Which induction motor has the highest torque
    a) Capacitor start
    b) Squirrel pole
    c) Shaded pole
    d) Split phase
20. The basic function of a transformer is to change
    a) Level of voltage
    b) The power level
    c) The power factor
    d) The frequency
21. Which of the following d.c generators has the poorest voltage regulation?
    a) Over-compounded generator
    b) Series generator
    c) Flat-compounded generator
    d) Shunt generator
22. What will be the total voltage drop at the brushes of a four pole dc motor
    a) 4
    b) 2
    c) 8
    d) 6
23. The speed of a d.c motor can be controlled by changing
    a) Its flux
    b) Armature circuit resistance
    c) Applied voltage
    d) All of the above
24. The magnetic flux in the core of a single phase transformer is
    a) Purely resistive one
    b) Purely alternating one
    c) Partly alternating and partly rotating one
    d) Purely inductive one

25. The approximate efficiency of an ordinary transformer is _____
   a) 65%
   b) 80%
   c) 50%
   d) 95%
26. The maximum voltage is induced in the rotor of a 3-phase induction motor when it_____
   a) Runs at full load
   b) Is blocked
   c) Runs at no load
   d) None of the above
27. The condition of an induction motor on no-load resemble those of transformer whose secondary is _____
   a) Open circuited
   b) Short circuited
   c) Overloaded
   d) Supplying a variable resistive load
28. The single phase series motor can operate on_____
   a) AC only
   b) DC only
   c) Unit power factor
   d) Both ac and dc
29. Majority of the alternators in use have_____
   a) Revolving a.c armature winding
   b) Stationery field type construction
   c) Revolving field type construction
   d) Revolving armature and revolving field
30. Which of the following formula is wrong?
   a) $E_g = V + I_a R_a$
   b) $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$
   c) $E = \frac{P\emptyset Zn}{60A}$
   d) $E_b = V_s + I_a R_a$
31. The efficiency of a DC generator means its
   a) Electrical efficiency
   b) Overall efficiency
   c) Mechanical efficiency
   d) Frictionless efficiency
32. How can the armature reaction be increased?
   a) When the field current increases
   b) When the armature current decreased
   c) When the armature current increases
   d) When the field current decrease

33. A separately excited d.c generator is not used because
    a) It is costly
    b) Voltage drop considerably with loud
    c) Separate d.c source required for field circuit
    d) One power source required for both the armature and field coils
34. The main purpose of using motor starter in dc motor is to
    a) Avoid excessive starting current
    b) Prevent fuse blowing
    c) Accelerate the motor
    d) Starting the motor
35. Which of the following statements is not true about transformers?
    a) Transformers step down AC voltage
    b) Transformers step up d.c voltages
    c) Transformers are inductors
    d) Transformers step up A.C voltages
36. One of the following is not true about motors which one is it
    a) Motors use Flemings right hand rule
    b) Motors use Flemings left hand rule
    c) Motors use mutual induction principle
    d) Motors use magnetic interaction principle
37. What will happen if the air-gap between the rotor and stator of an ac induction motor is increased
    a) The motor will stall
    b) The magnetic field will increase
    c) The magnetic field reluctance will improve
    d) The will be high reluctance to the flow of magnetic flux
38. The main reason of running a high powered motor in star-delta connection is to
    a) Reduce number of conductors supplying power to the motor
    b) Reduce the speed of the motor
    c) Increase the efficiency of the motor
    d) Reduce the starting current
39. Which one is not a factor when calculating real power in a three phase motor?
    a) Applied voltage
    b) Sine of angle theta
    c) Power factor
    d) Cosine of angle theta
40. Which of the following single phase motors has the lowest torque?
    a) Capacitor start motor
    b) Capacitor run motor
    c) Shaded pole motor
    d) Reactance start motor

SECTION 2: SHORT ANSWER QUESTIONS

1. a) State two methods of connecting armature coils to the commutator segments (2marks)

b) Name any three types of d.c generators. (3marks)

2. a) State three functions of a commutator in a d.c motor. (3marks)

b) State two methods used to improve commutation (2marks)

3. a) What are the four factors that can be controlled in the operation of a d.c motor? (4marks)

b) What is commutation (1 mark)

4. State three disadvantages of 3-phase transformers and two disadvantages of a three-phase transformers over a single –phase transformer (5marks)
(2 marks)

5. A 230V series motor draws 50A from the supply. The resistances of the armature and series field windings are 0.2Ω and 0.1Ω respectively. Calculate:
   a) Brush voltage (1mark)
   b) Back EMF (2mark)
   c) Power wasted in the armature (2marks)

6. Calculate the speed and open-circuit line and phase voltages of a 4pole 3phase 50Hz star connected alternator with 36 slots and 30 conductors per slot. The flux per pole is 0.05wb and winding factor is 1 (5marks)

7. ii) Name the two types of rotors used in induction motors (2marks)

ii) A 4 pole 3phase induction motor operates from a supply whose frequency is 50Hz. Calculate a synchronous speed with which the magnetic field of the stator is rotating and the speed of the rotor. (3marks)

8. i) A transformer is the most efficient electrical machine. What are the main loses in a transformer? (3marks)
   ii) Name any two types of transformers? (2marks)

SECTION 3: LONG ANSWER QUESTIONS

1.
   i. State and describe five parts of a DC machine. (10marks)

   ii. Draw and label the diagram of capacitor start capacitor run single phase induction motor (5marks)

   iii. State five applications where a capacitor start capacitor run single phase induction motor are used. (5marks)